

The Limbic System

Sitting below the cortex and above the brain stem, the limbic system is a collection of structures associated with emotion, memory, and basic instincts.

Location and function

The limbic system is a cluster of organs situated in the center of the brain, occupying parts of the medial surfaces of the cerebral cortex. Its major structures form a group of modules that pass signals between the cortex and the bodies of the lower brain. Nerve axons link all of its parts and connect them to other brain areas. The limbic system mediates instinctive drives such as aggression, fear, and appetite, with learning, memory, and higher mental activities.

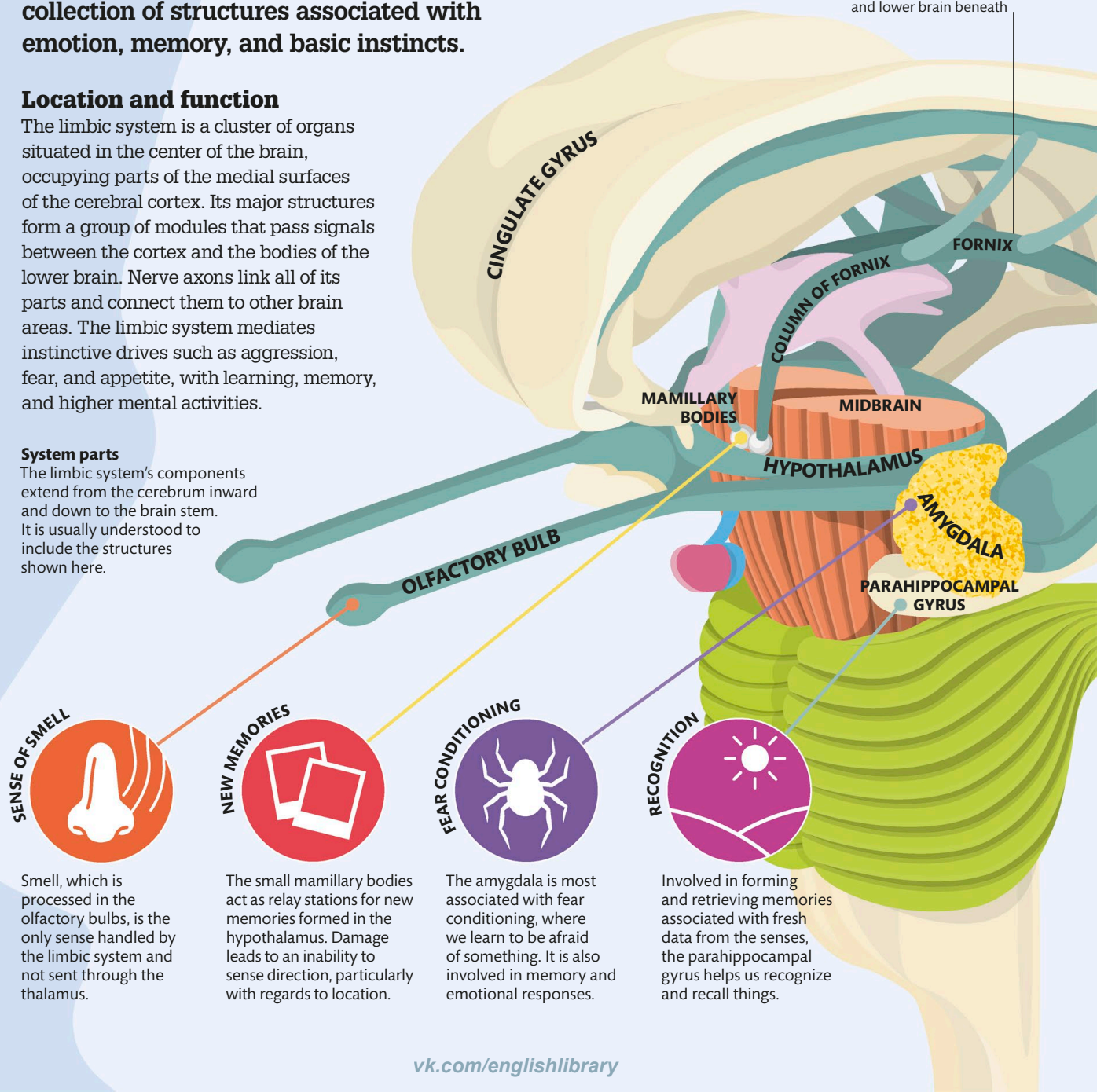
System parts

The limbic system's components extend from the cerebrum inward and down to the brain stem. It is usually understood to include the structures shown here.

THE S-SHAPED
HIPPOCAMPUS IS NAMED
AFTER ITS RESEMBLANCE
TO A **SEAHORSE**



Fornix is a bundle of nerve tracts that connects hippocampus to thalamus and lower brain beneath



SENSE OF SMELL



Smell, which is processed in the olfactory bulbs, is the only sense handled by the limbic system and not sent through the thalamus.

NEW MEMORIES



The small mamillary bodies act as relay stations for new memories formed in the hypothalamus. Damage leads to an inability to sense direction, particularly with regards to location.

FEAR CONDITIONING



The amygdala is most associated with fear conditioning, where we learn to be afraid of something. It is also involved in memory and emotional responses.

RECOGNITION



Involved in forming and retrieving memories associated with fresh data from the senses, the parahippocampal gyrus helps us recognize and recall things.